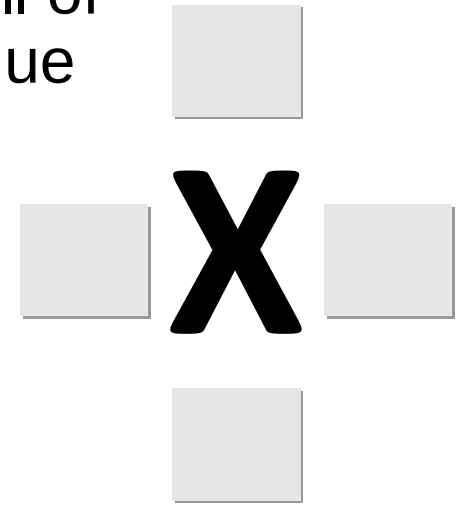
CHAPTER 8

Bonding and Molecular Structure

Lewis Symbols

- Valence e^-
  - e^- from the shell of the highest n value
- Octet rule
  - noble gas e^- configuration is stable
    - $ns^2\,np^6$
- Useful only for “A” group elements



Basic rules

Draw the atomic symbol. This represents the atomic nucleus along with the core elements.

Treat each side as a box that can hold up to two electrons.

Count the electrons in the valence shell.

Start filling boxes - don’t make pairs unless you need to.

Lewis Structures

- : & —
  - Cl:Cl or Cl — Cl
- lp (lone pair)
- bp (bonding pair)
- octet rule/duet rule for H
- Examples
  - H_2
  - HF

- The electron pair bond between the two atoms of an H_2 molecule is represented by a pair of dots or a line.

H atom:

H·

H_2 molecule:

H:H

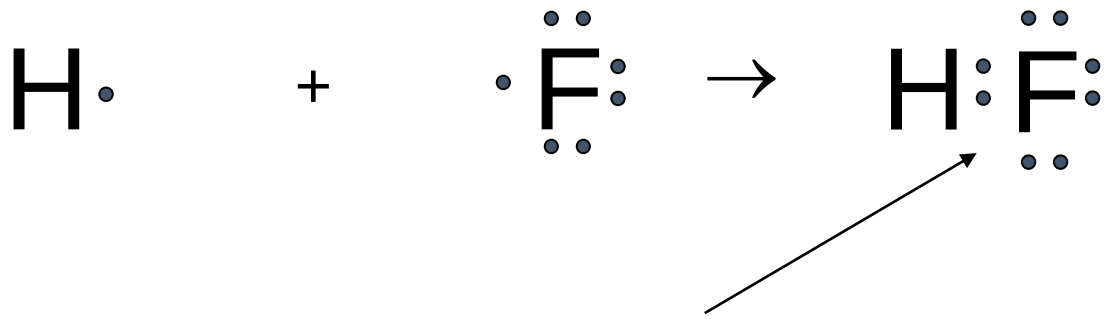
H–H

- The representation of a molecule in this fashion is called a **Lewis electron dot structure** or just a **Lewis structure** in honor of G. N. Lewis.

Lewis Dot Structures & the Octet Rule

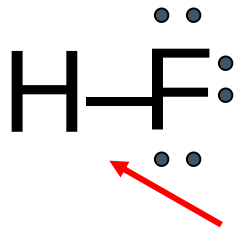
When other covalent species form, there are additional electron pairs that do not participate in bonding.

These are called "***lone pairs***" (*lp*)



one *bonding pair* & three *lone pairs* (octet)

hydrogen fluoride: HF



"single bond"

Covalency

The number of bonds an atom normally forms

How many e^- from other atoms are needed to form an octet?

Atom	H	Cl	O	N	C
Valence	1	7	6	5	4
Covalency	1	1	2	3	4

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Writing Lewis Structures

1 Calculate the total number of valence e^-

- anions - add one e^- for each negative charge.
- cations - subtract one e^- for each positive charge.

2 Write the skeleton structure. (In general, the least electronegative atom occupies the central position; hydrogen and fluorine occupy terminal positions.)

3 Place one pair of e^- between bonded atoms.

4 Complete the octets of the atoms attached to the central atom by adding electron pairs.

5 Count the total number of e^- ; subtract from Step 1

- Place any remaining e^- on the central atom in pairs.
- If the central atom has fewer e^- than an octet, form double or triple bonds as necessary.

** Hydrogen atoms will never have lone pairs or multiple bonds

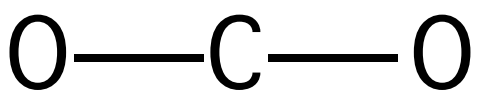
Multiple Bonds

- Acetylene, C_2H_2
- Nitrogen, N_2

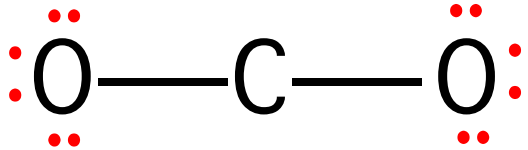
- Atoms that have a tendency to form multiple bonds:
  - C, O, N, S, P

Carbon Dioxide, CO_2

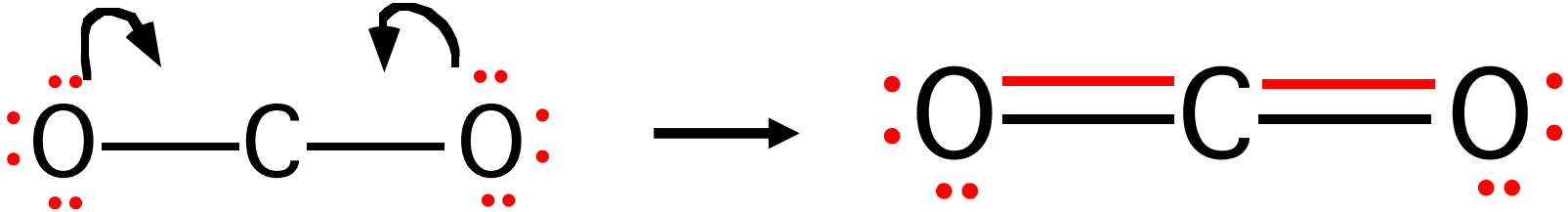
1. Central atom = C
2. Valence electrons = 16 valence electrons
3. Form bonds.



4. Place lone pairs on outer atoms.



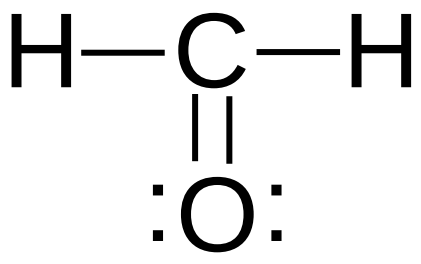
5. So that C has an octet, we shall form DOUBLE BONDS between C and O.



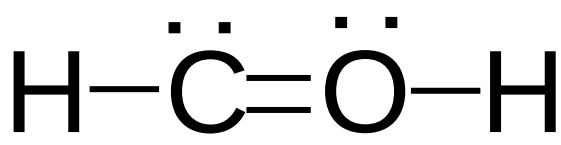
The second bonding pair forms a *double* bond.

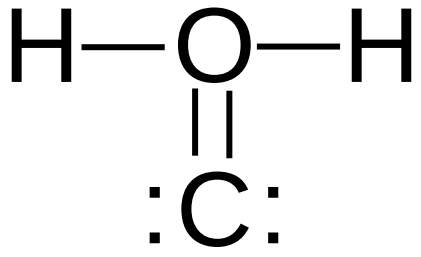
Formaldehyde, CH_2O

Possible Structures:



Which is more stable?





Using Formal Charge to Predict Stability

- When several Lewis structures are possible, the most stable structure will be when
  - the atoms have the smallest amount of formal charge
  - any negative charge resides on the more electronegative atom

F.C. = number of valence e^- – [lone pair e^- + ½ bonding e^-]

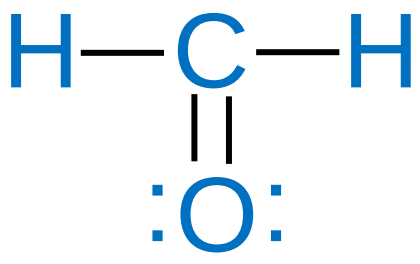
More on Formal Charge

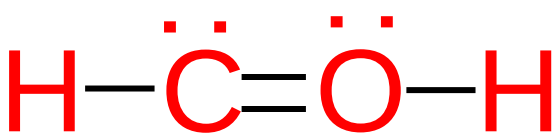
- Where formal charges are required, they should be as small as possible, and negative formal charges should appear on the most electronegative atoms.
- Adjacent atoms in a structure should not carry formal charges of the same sign.
- Formal charges on the atoms in a Lewis structure must total to zero for a molecule and to the net charge for a polyatomic ion.
- The sum of the atoms in an ion equals the charge on the ion
- Usually, the most plausible Lewis structure is the one with no formal charges (formal charges of zero on all atoms).

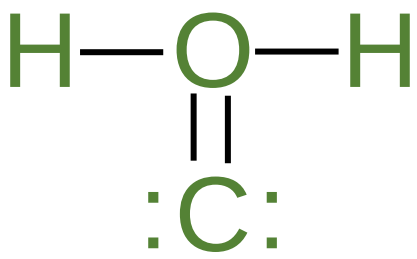
Which is the correct Lewis structure of Formaldehyde?

F.C. = number of valence e– – [lone pair e– + ½ bonding e–]

- F.C => C: 4 – [0 + ½*8] = 0
  O: 6 – [4 + ½ *4] = 0
  H: 1 – [0 + ½*2] = 0
- F.C.=> C: 4 – [2+ ½*6] = -1
  O: 6 – [2 + ½* 6] = 1
  H: 1 – [0 + ½* 2]=0
- F.C.=> C: 4 – [4 + ½ *4] =-2
  O: 6 – [0 + ½* 8] = 2
  H: 1 – [0 – ½*2] = 0







Write the Lewis Structure and Determine the Formal Charge of HCN and HNC

- Which is the best way to represent hydrocyanic acid?
- Why?

HCN

- H – C ≡ N :
- F.C. (H) = 1 – [0+1/2*(2)] =0
- F.C. (C) = 4 – [0+1/2*(8)]=0
- F.C. (N) = 5 – [2+1/2*(6)]=0

Sum of formal charges are zero!

HNC

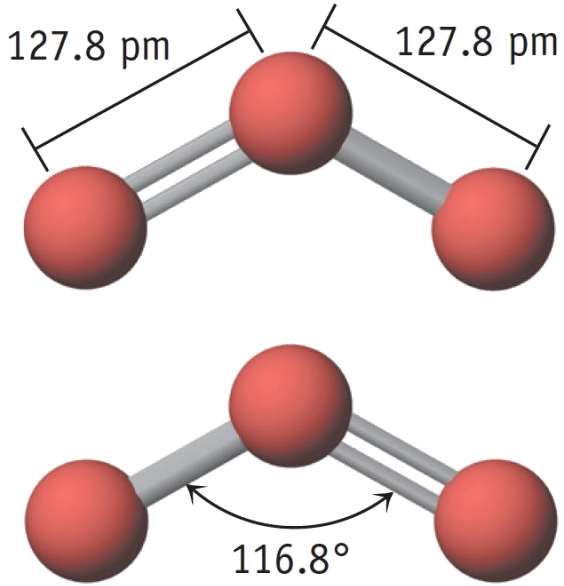
- $H - N^{+} \equiv C^{-}$:
- F.C. (H) = 1 – [0+1/2*(2)] =0
- F.C. (N) = 5 – [0+1/2x(8)]=+1
- F.C. (C) = 4 – [2+1/2(6)]=-1

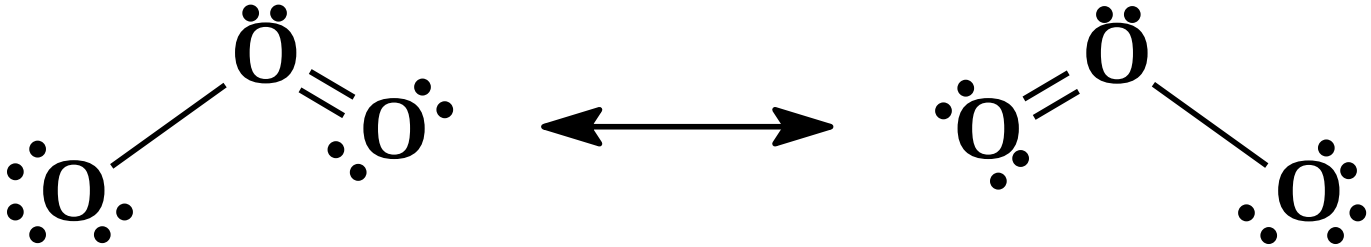
Sum of formal charges are zero!

Resonance Structures

- Equivalent Lewis structures that are the same except for the placement of e^-
  - more than one Lewis structure
  - structure contains double or triple bonds
  - only one bond length exists
  - bond length is shorter than a single bond but not as short as a double or triple bond
- Resonance structures have the same arrangements of atoms but different arrangements of electrons
- The actual structure of the molecule is a composite, or hybrid, of the equivalent resonance structures

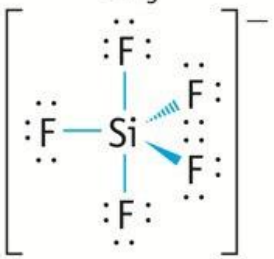
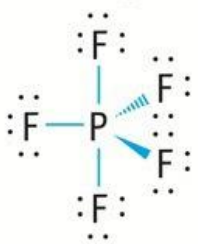
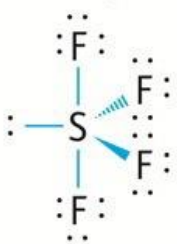
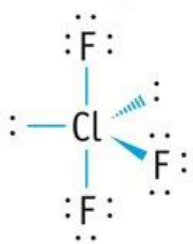
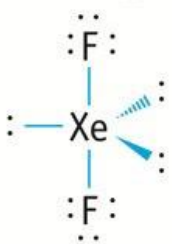
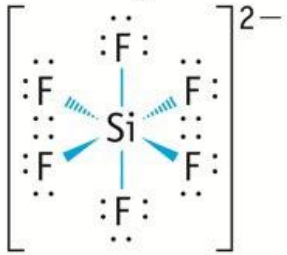
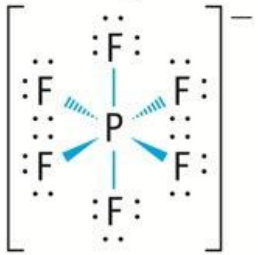
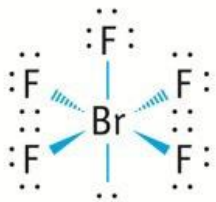
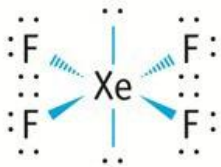
Resonance Structures





Compounds in Which an Atom Has More Than Eight Valence Electrons

Table 8.5 Lewis Structures for Hypervalent Compounds in Which the Central Atom Exceeds an Octet

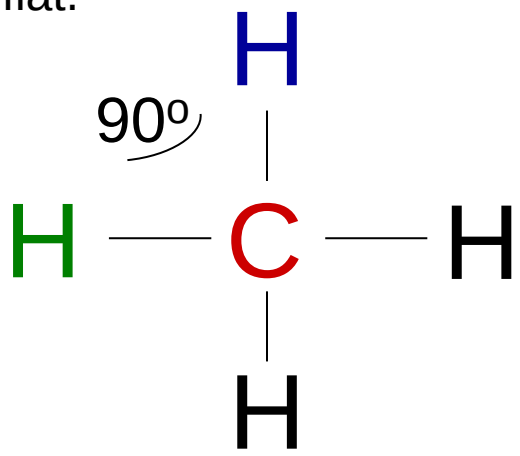
Group 4A	Group 5A	Group 6A	Group 7A	Group 8
SiF_5^- 	PF_5 	SF_4 	ClF_3 	XeF_2 
SiF_6^{2-} 	PF_6^- 	SF_6	BrF_5 	XeF_4 

Molecular Shape

Lewis structures tell us **how atoms are connected** in a molecule:
bonds *(bp)* & lone pairs (*lp)* etc…

The 3-D shape of a molecule is however determined by its bond angles.

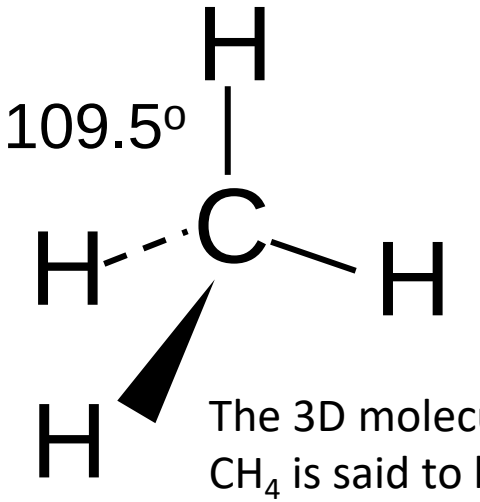
The Lewis structure suggests that the H–C–H bond angles are 90º & that the molecule is flat.



The methane molecule must be 3-D since the sum of the angles is greater than 360 degrees!

Experimentally however, the H–C–H bond angles are found to be 109.5°

If this is the case, the molecule cannot be planar…



The 3D molecular geometry of CH_4 is said to be "***tetrahedral***"

Molecular Shape

- In order to predict molecular shape, we assume the valence electrons of each atom in the molecule repel one another.
- When this occurs, the molecule adopts a 3-D geometry that minimizes this repulsion where:

lp-lp > lp-bp > bp-bp

This process is known as:

Valence Shell Electron Pair Repulsion (VSEPR) theory.

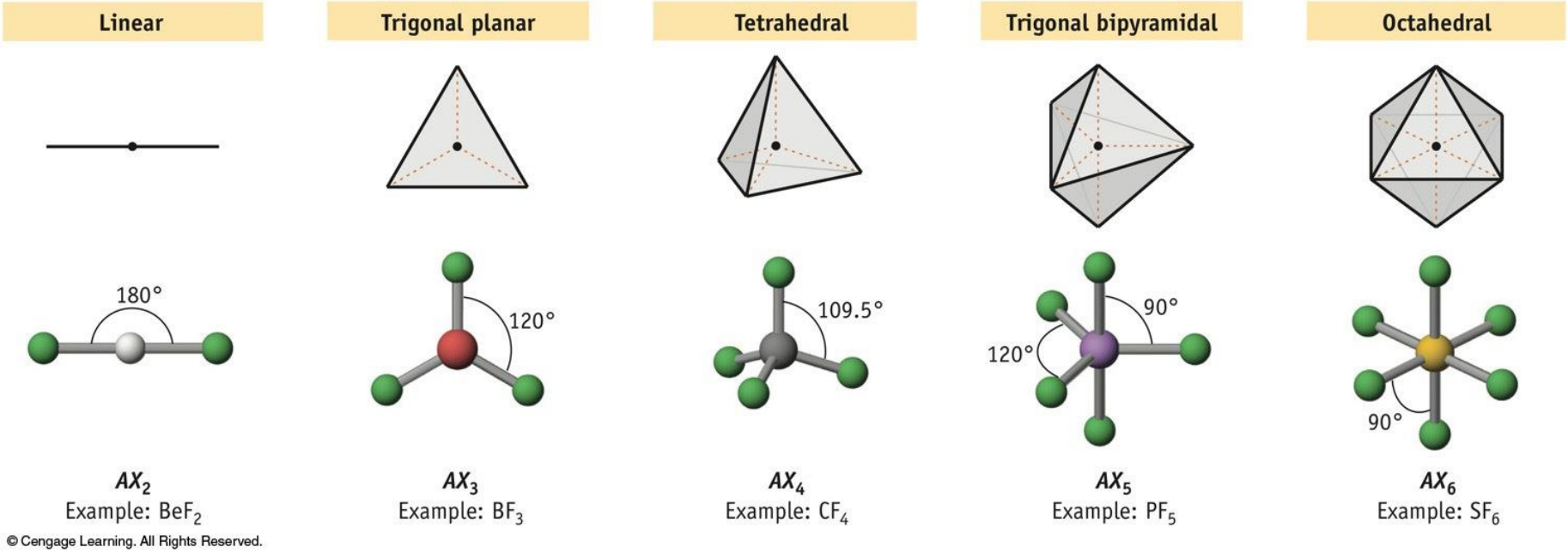
Geometry

- **Molecular** geometry
  - Arrangement of atoms in space

- **Electron-Domain** geometry
  - Arrangement of electron pairs in space

- **Molecular** geometry = **Electron-Domain** geometry
  - When there are no lone pair in the central element

Molecular and Electron-Pair Geometries



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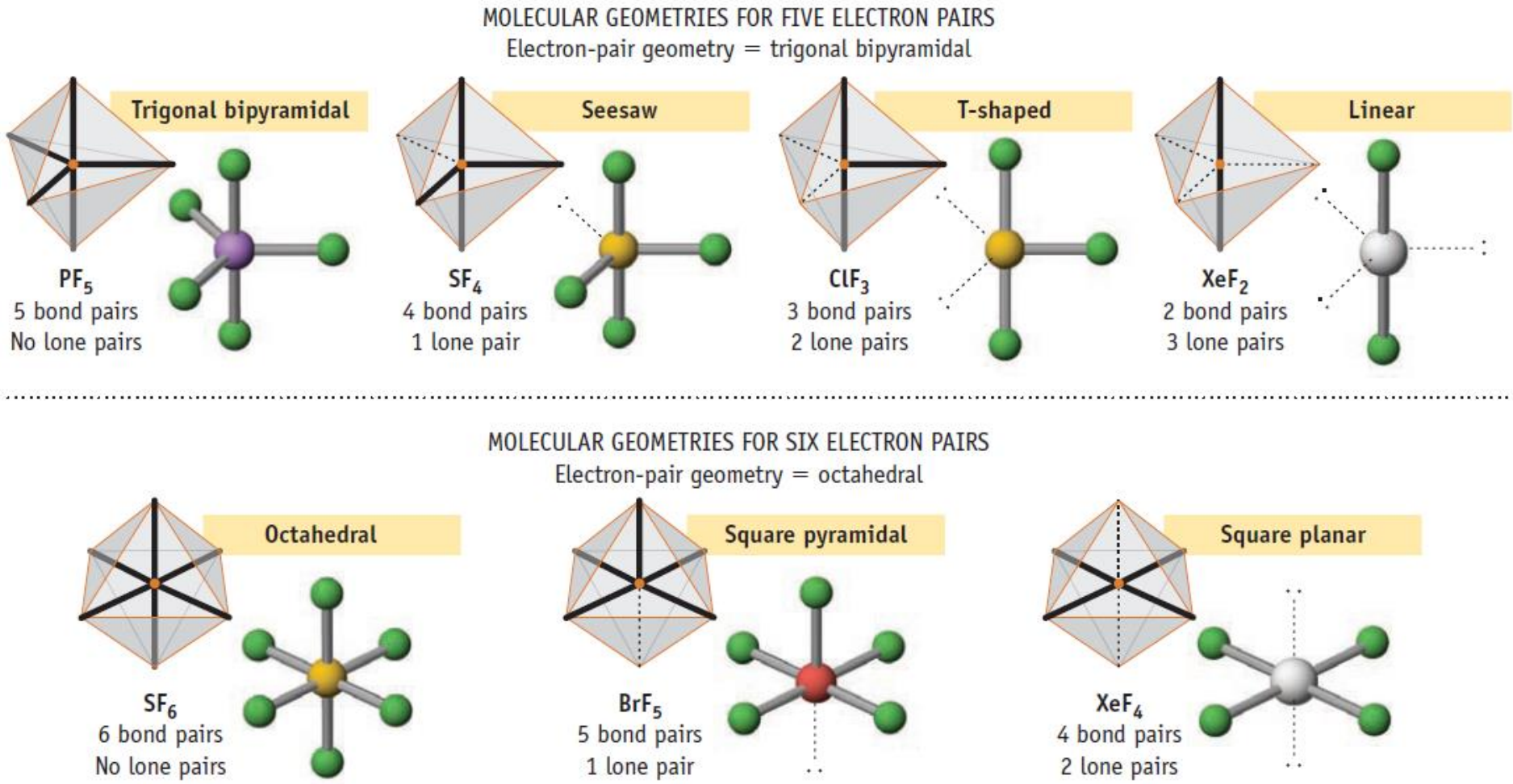
2 pairs of electrons	3 pairs of electrons	4 pairs of electrons	5 pairs of electrons	6 pairs of electrons

Central Atoms Surrounded Only by Single-Bond Pairs

Molecular Geometries for 5 Electron Pairs



Electron domain Geometry

Linear | Trigonal planar | Tetrahedral | Trigonal bipyramidal | Octahedral

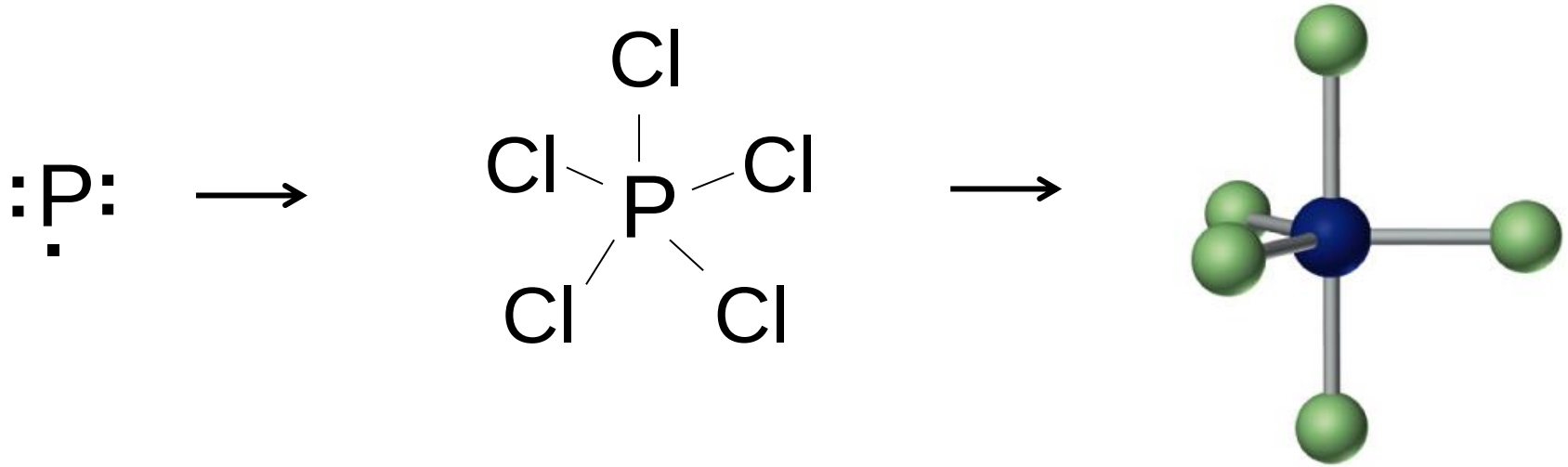
Lone pairs count as an electron domain
Double and triple bonds count as only one electron domain

6 pairs - octahedral
5 pairs - trigonal bipyramidal
4 pairs - tetrahedral
3 pairs - trigonal planar
2 pairs - linear

Example: PCl_5

P has 5 valence electrons

Each Cl atom needs one electron to complete the octet



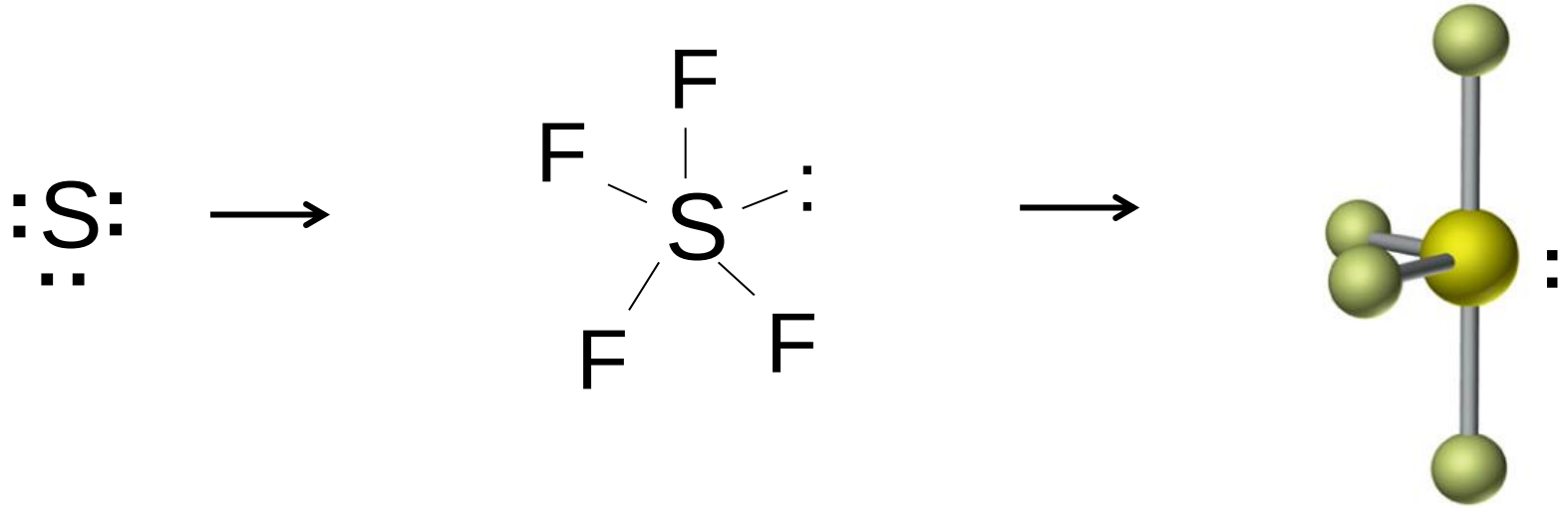
Molecular Geometry : *Trigonal Bipyramidal*

Electron-Pair Geometry: *Trigonal Bipyramidal*

Example: SF_4

S has 6 valence electrons

Each F atom needs one electron to complete the octet

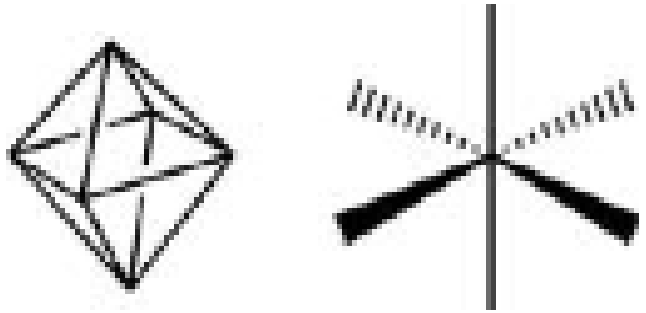


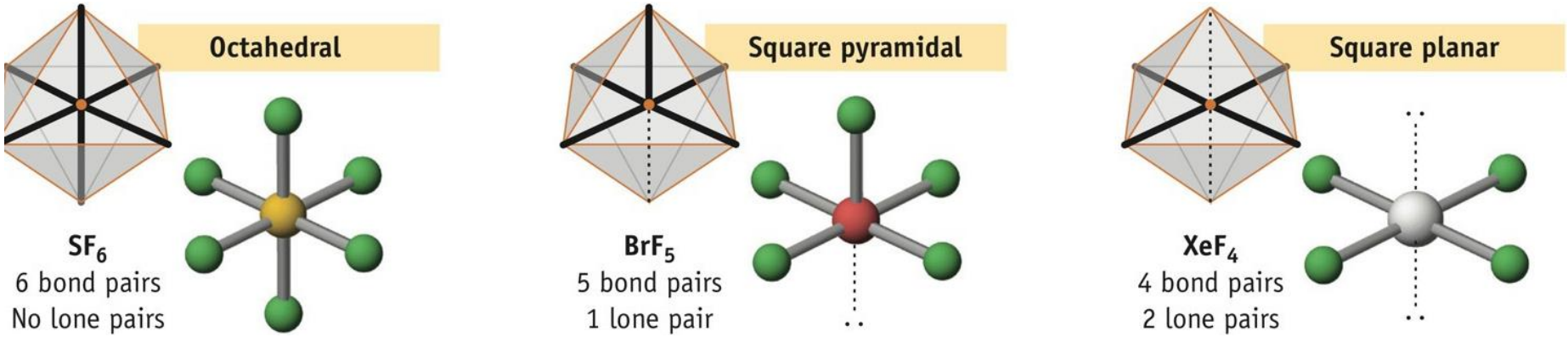
Molecular Geometry : *Seesaw*

Electron Pair Geometry: *Trigonal Bipyramidal*

Molecular Geometries for 6 Electron Pairs

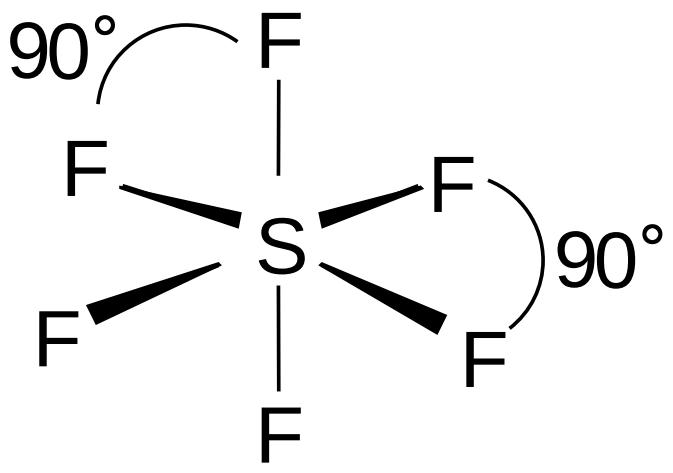
All are based on the 8-sided octahedron

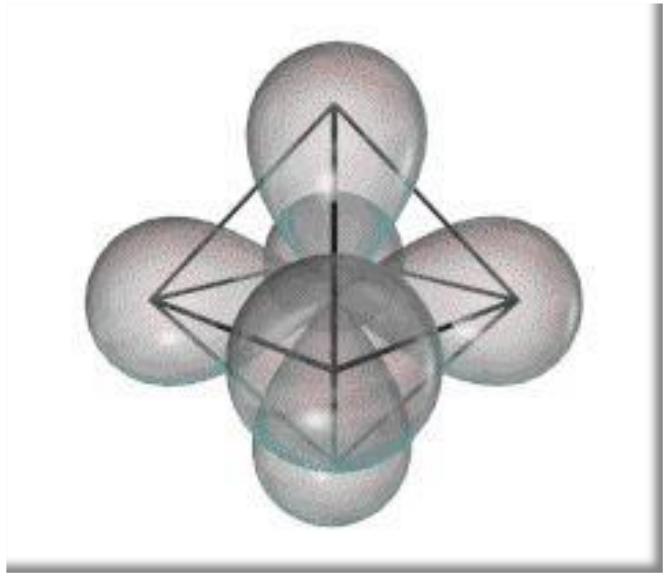




Compounds with 5 or 6 Pairs Around the Central Atom

Octahedron



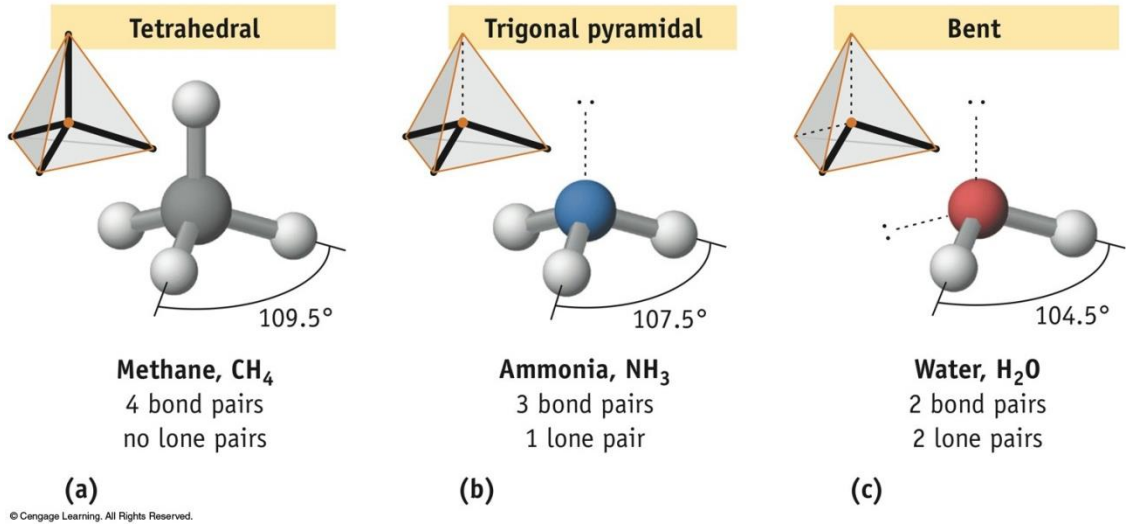


6 electron pairs

Predicting Molecular (or Ion) Shape

1. Write the Lewis structure
2. Count the number of electron domains around the central atom to determine the arrangement minimizing repulsion
3. Describe the molecular geometry in terms of the arrangement of the bonding domains.
4. Double and triple bonds are counted as *one domain* for determining molecular shape
5. For the VSEPR model, multiple bonds count as one effective electron pair.

Influence of Lone Pairs on the Molecular Geometry

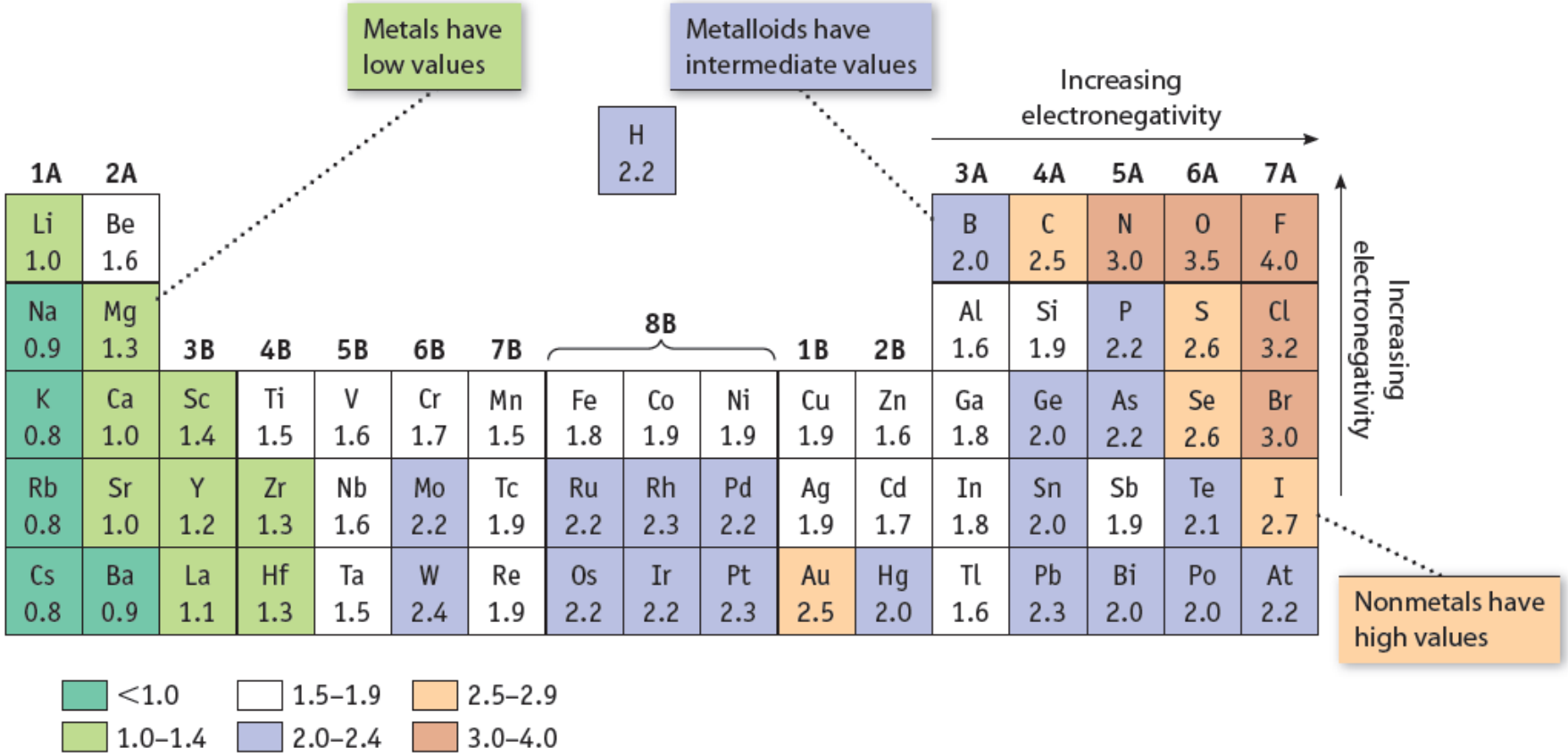


Lone pairs require more room than bonding pairs and tend to compress the angles between the bonding pairs.

Bond angles:

CH_4	109.5	0 LP	Tetrahedral
NH_3	107.5	1 LP	Trigonal pyramidal
H_2O	104.5	2 LP	Bent

Electronegativity and Bond Polarity

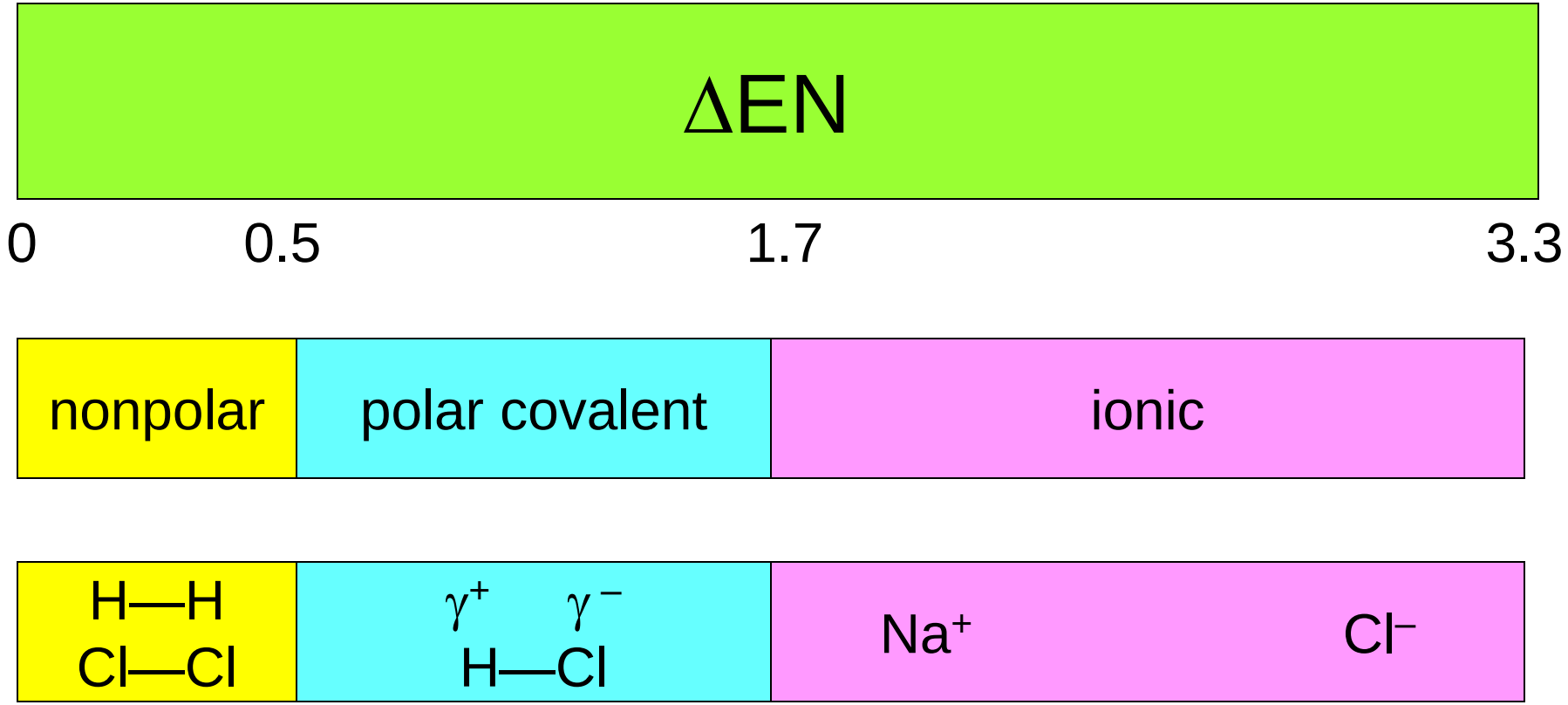


Recall that electronegativity (χ) measures the relative tendency for an atom to polarize a bond. It follows Z_{eff} on the periodic table.

Electronegativity

- Electronegativity tells us how much a particular atom “wants” electrons
- Electronegativity increases (left to right) across a period
- Electronegativity decreases down the periodic table
- A molecule with no polar bonds is nonpolar molecule
- A molecule with 1 polar bond is a polar molecule
- If the individual bond dipoles do not cancel, the molecule is polar
- If the individual bond dipoles cancel, the molecule is nonpolar
- If there are lone pairs on the central atom, the molecule is polar

Electronegativity Difference

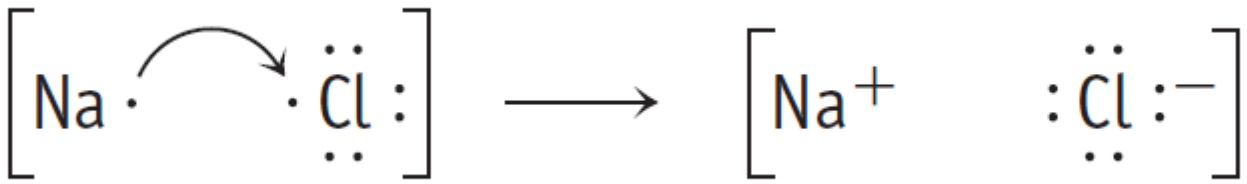


The Chemical Bond

- The force that holds two atoms together
  - Bonding occurs using valence electrons
  - Electrons in an atom can be categorized as either valence or core electrons
  - Atoms try to get eight electrons in the valence
  - Lewis symbols help us to track the valence electrons and predict bonds.

Ionic Bonding

complete transfer of 1 or more electrons



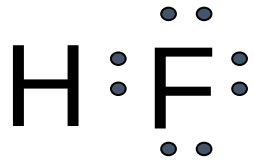
electron transfer
from reducing agent
to oxidizing agent

ionic compound.
Ions have noble gas
electron configurations.

Covalent bonds

- e^- pair sharing
  - Valence e^- (outer shell electrons)
  - Nonmetal—nonmetal bonds

*The presence of a metal in a formula is a signal that the compound is likely Ionic



Polar Covalent Bonds

- Electrons may not be shared equally.

- This is based on the electronegativity difference between the two elements forming the bond.

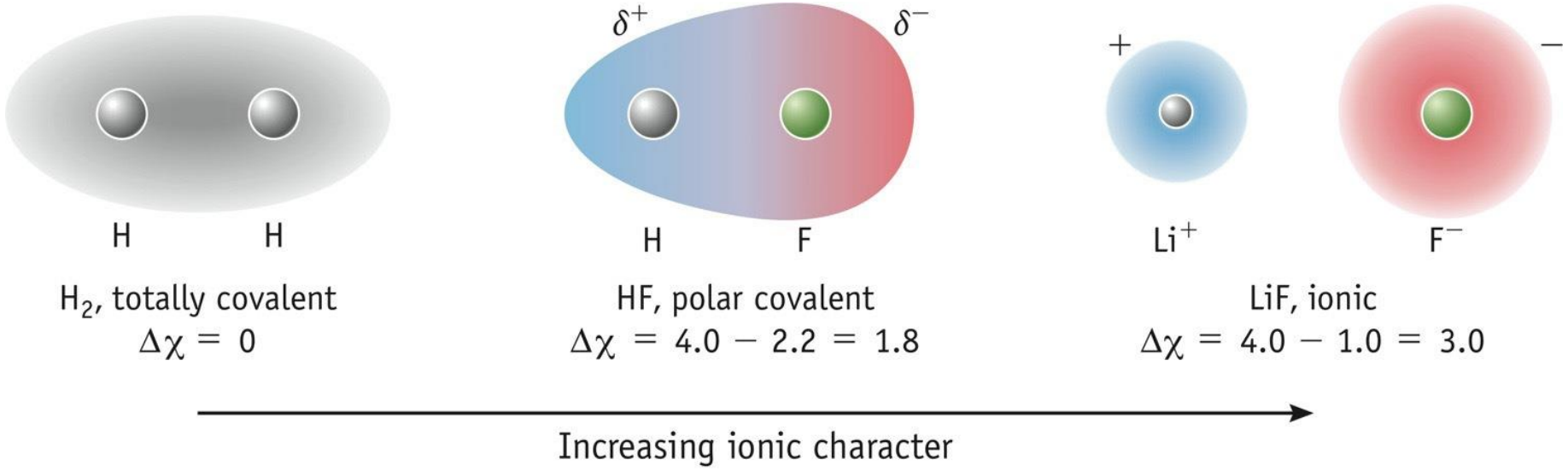
- If difference = 0 covalent
- difference > 1.7 ionic

- Values between 0 and 1.7 are polar covalent.
  - - electrons are not shared equally.

Bond Polarity & Electronegativity

As the difference in Electronegativity increases ($\Delta\chi$), so does the ionic character of the bond.

Increasing covalent character



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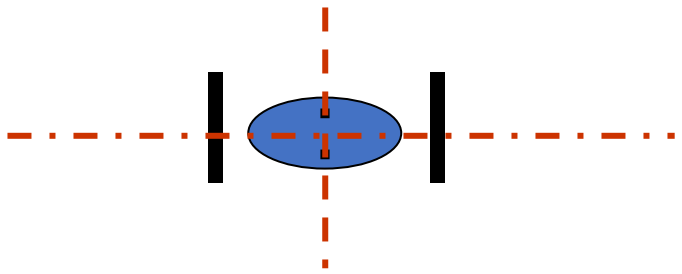
Bond Polarity & Electronegativity

- A difference in electronegativity of 0 indicates that the bond is *purely covalent*. There is equal sharing of electrons.
- A difference of electronegativity between 0.1 to 0.3 indicates that the bond is *nonpolar covalent*.
- A difference of electronegativity between 0.4 to 1.7 indicates that the bond is *polar covalent*.
- A difference of electronegativity that is greater than 1.8 indicates that the bond is *ionic*.

Overall Polarity in Molecules

The polarity of the individual bonds in a molecule will determine the overall polarity of a molecule.

All *homonuclear* diatomic molecules are non-polar.



iodine (I_2)

$\Delta\chi = 2.7 - 2.7 = 0$

Notice the symmetry of the molecule:

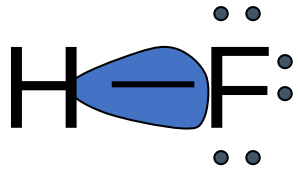
- When divided, the top and bottom as well as the left and right are mirror images of one another.
- One also knows the molecule is nonpolar because the bond is nonpolar.

Example: HF

- Fluorine has a larger electronegativity value than hydrogen.
- This means that the electrons in the bond are skewed toward the F atom.
- The electrons shift toward the F atom.

	3A	4A	5A	6A	7A
H 2.2	B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
	Al 1.6	Si 1.9	P 2.2	S 2.6	Cl 3.2
	Ga 1.8	Ge 2.0	As 2.2	Se 2.6	Br 3.0
	In 1.8	Sn 2.0	Sb 1.9	Te 2.1	I 2.7
	Tl 1.6	Pb 2.3	Bi 2.0	Po 2.0	At 2.2

This polarizes the molecule



more positive end (+)

more negative end (–)

Electronegativity: H vs. F

The greater the difference in electronegativity ($\Delta\chi$) the more polar the covalent bond.

H vs. F

4.0 – 2.2 = 1.8

C vs. F

4.0 – 2.5 = 1.5

O vs. F

4.0 – 3.5 = 0.5

H 2.2				
3A	4A	5A	6A	7A
B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
Al 1.6	Si 1.9	P 2.2	S 2.6	Cl 3.2
Ga 1.8	Ge 2.0	As 2.2	Se 2.6	Br 3.0
In 1.8	Sn 2.0	Sb 1.9	Te 2.1	I 2.7
Tl 1.6	Pb 2.3	Bi 2.0	Po 2.0	At 2.2

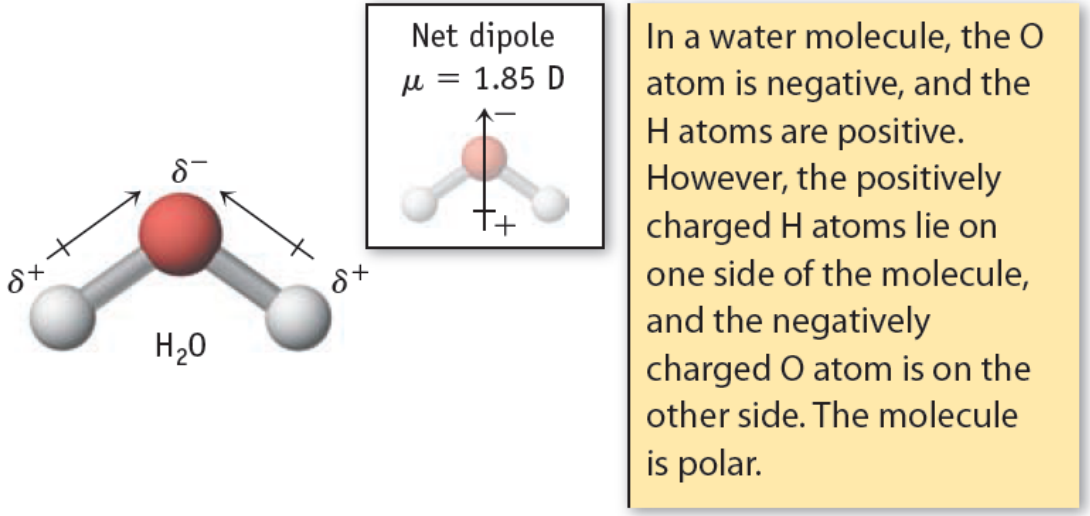
So, in terms of polarity,

H-F > C-F > O-F

Bond Polarity & Molecular Polarity

When a molecule possesses a net dipole moment, it is polar.

The individual O-H bond dipoles result in a “net” or overall dipole moment for the molecule.



In a water molecule, the O atom is negative, and the H atoms are positive. However, the positively charged H atoms lie on one side of the molecule, and the negatively charged O atom is on the other side. The molecule is polar.

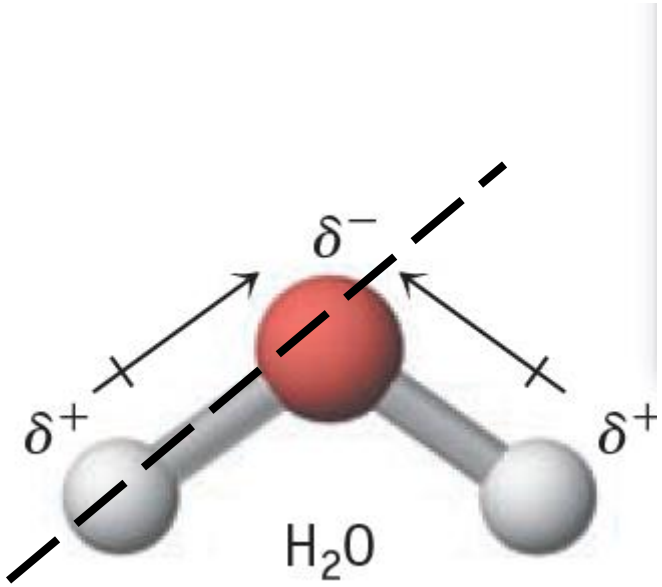
The molecule is therefore polar.

H_2O

Molecular Shape & Molecular Polarity

Notice also that the molecule's symmetry can be broken along either O-H bond axis:

This side..

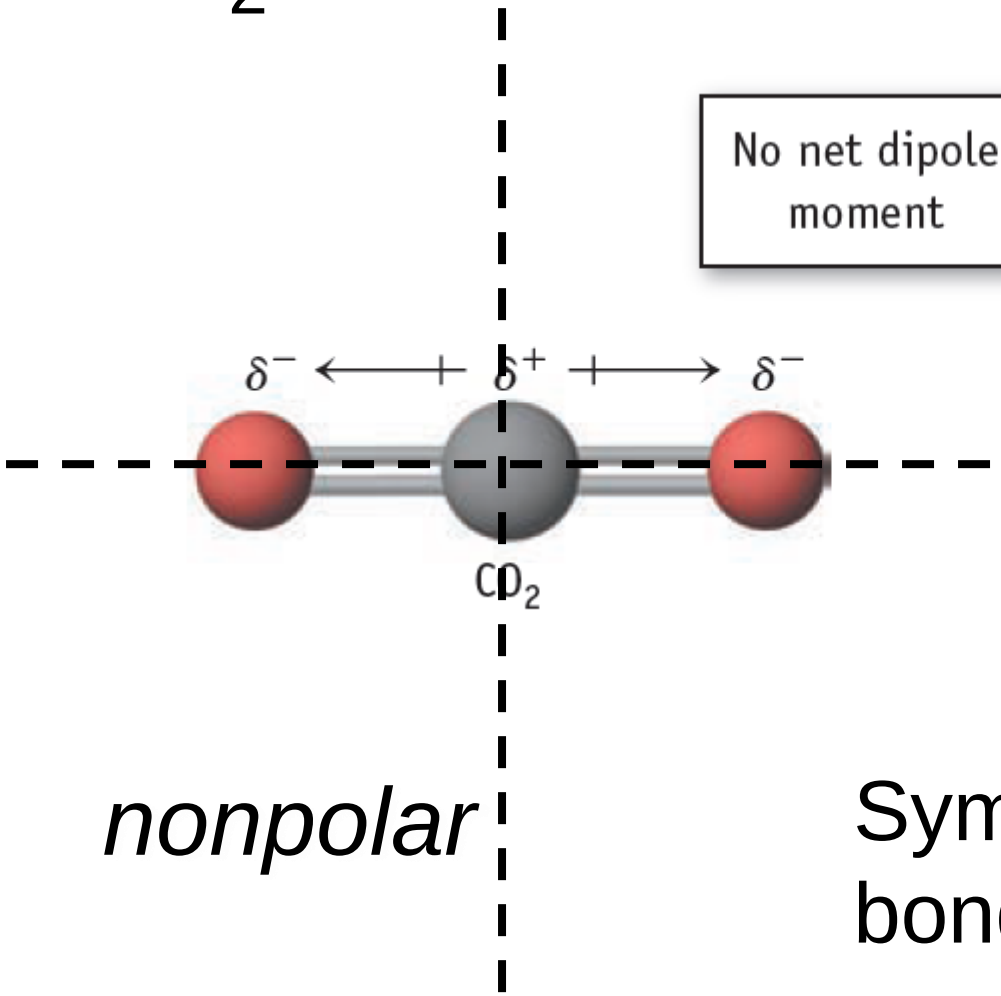


does not look like this side!

Whenever there exists a line or plane of asymmetry, the molecule is Polar!

Molecular Shape & Molecular Polarity

CO_2



No net dipole moment

For CO_2, the CO bonds are polar, but the electron density is distributed symmetrically over the molecule, and the charges of δ⁻ lie 180° apart. The molecule has no net dipole.

nonpolar

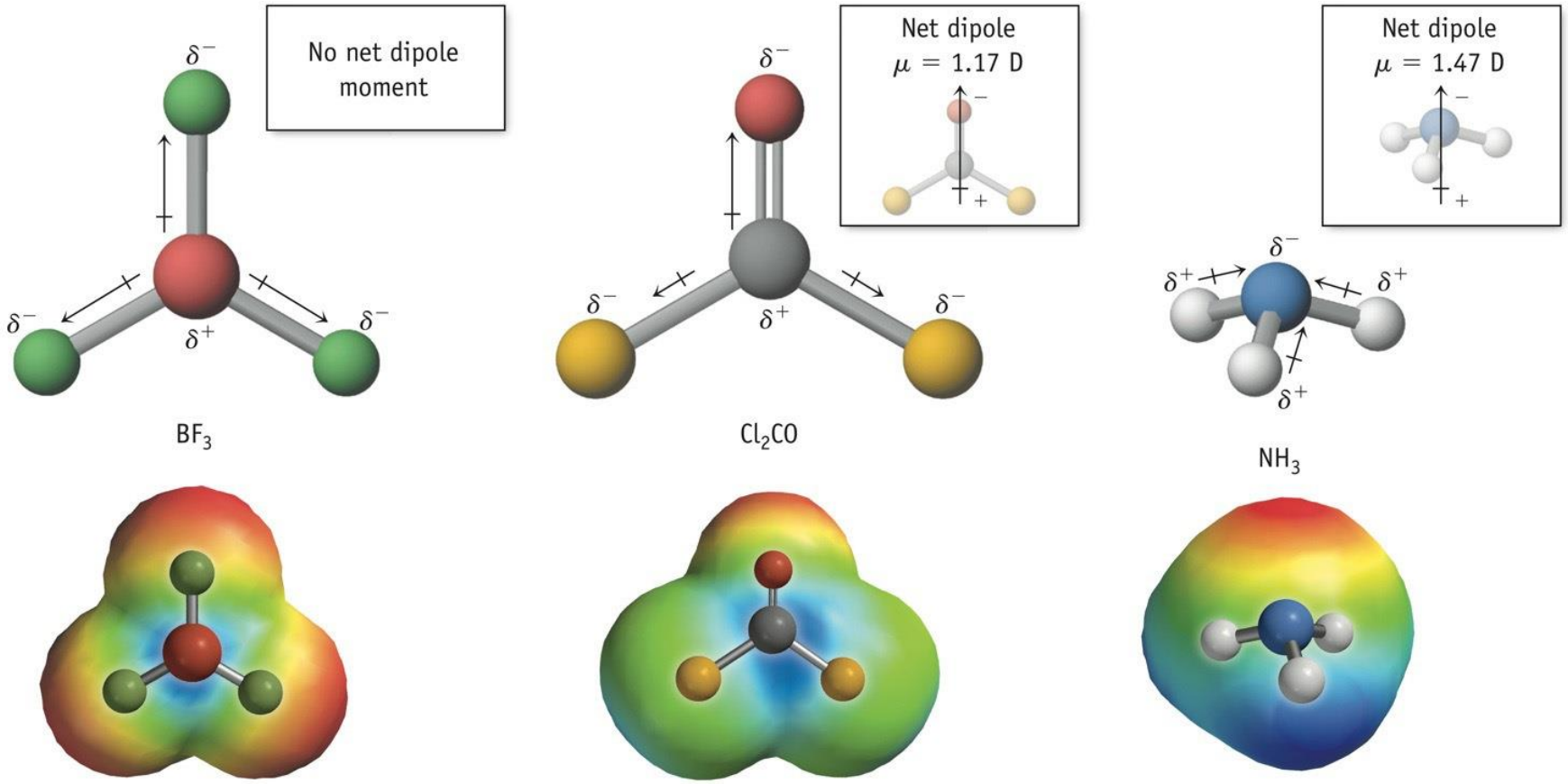
Symmetry across all bond axes.

Bond and Molecular Polarity

Table 8.6 Dipole Moments (μ) of Selected Molecules (in debye units)

Molecule (AX)	Moment (μ, D)	Geometry	Molecule (AX_2)	Moment (μ, D)	Geometry
HF	1.78	Linear	H_2O	1.85	Bent
HCl	1.07	Linear	H_2S	0.95	Bent
HBr	0.79	Linear	SO_2	1.62	Bent
HI	0.38	Linear	CO_2	0	Linear
H_2	0	Linear			
Molecule (AX_3)	**Moment (μ, D)**	**Geometry**	**Molecule (AX_4)**	**Moment (μ, D)**	**Geometry**
NH_3	1.47	Trigonal pyramidal	CH_4	0	Tetrahedral
NF_3	0.23	Trigonal pyramidal	CH_3Cl	1.92	Tetrahedral
BF_3	0	Trigonal planar	CH_2Cl_2	1.60	Tetrahedral
			$CHCl_3$	1.04	Tetrahedral
			CCl_4	0	Tetrahedral

Bond and Molecular Polarity



BF_3. The negative charge on the F atoms is distributed symmetrically, so the molecular dipole is zero.

In Cl_2CO and NH_3, the negative charge in the molecules is shifted to one side and the positive charge to the other side so the molecules are polar.

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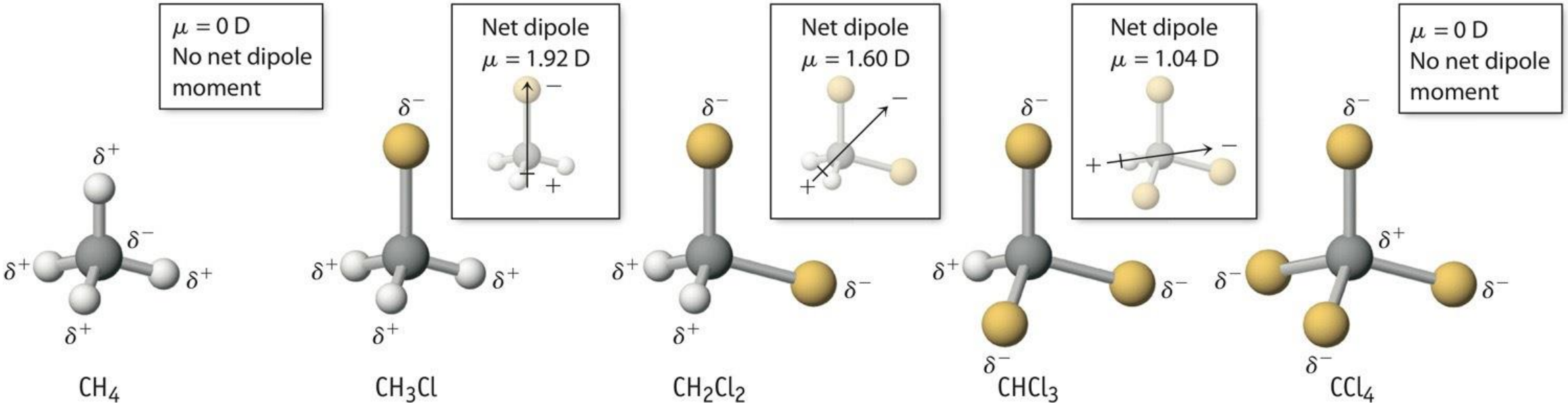
Dipole Moments & Molecular Polarity



Relationship between Molecular Geometry and Dipole Moment

Formula	Molecule Geometry	Dipole moment
AX	Linear	Can be 0
AX_2	Linear	Zero (same ligand)
	Bent	Can be 0
AX_3	Trigonal Planar	Zero (same ligand)
	Trigonal pyramidal	Can be 0
	T-shaped	Can be 0
AX_4	Tetrahedral	Zero (same ligand)
	Square planar	Zero (same ligand)
	See saw	Can be 0
AX_5	Trigonal planar	Zero (same ligand)
	Square planar	Can be 0
AX_6	Octahedral	Zero (same ligand)

Bond Properties: Order, Length, Energy

Covalent bond strength increases with increasing $\Delta\chi$

example: HCl bond is stronger than the HBr bond

Covalent bond strength increases with increasing bond order.

example: O=O bond is stronger than O–O bond

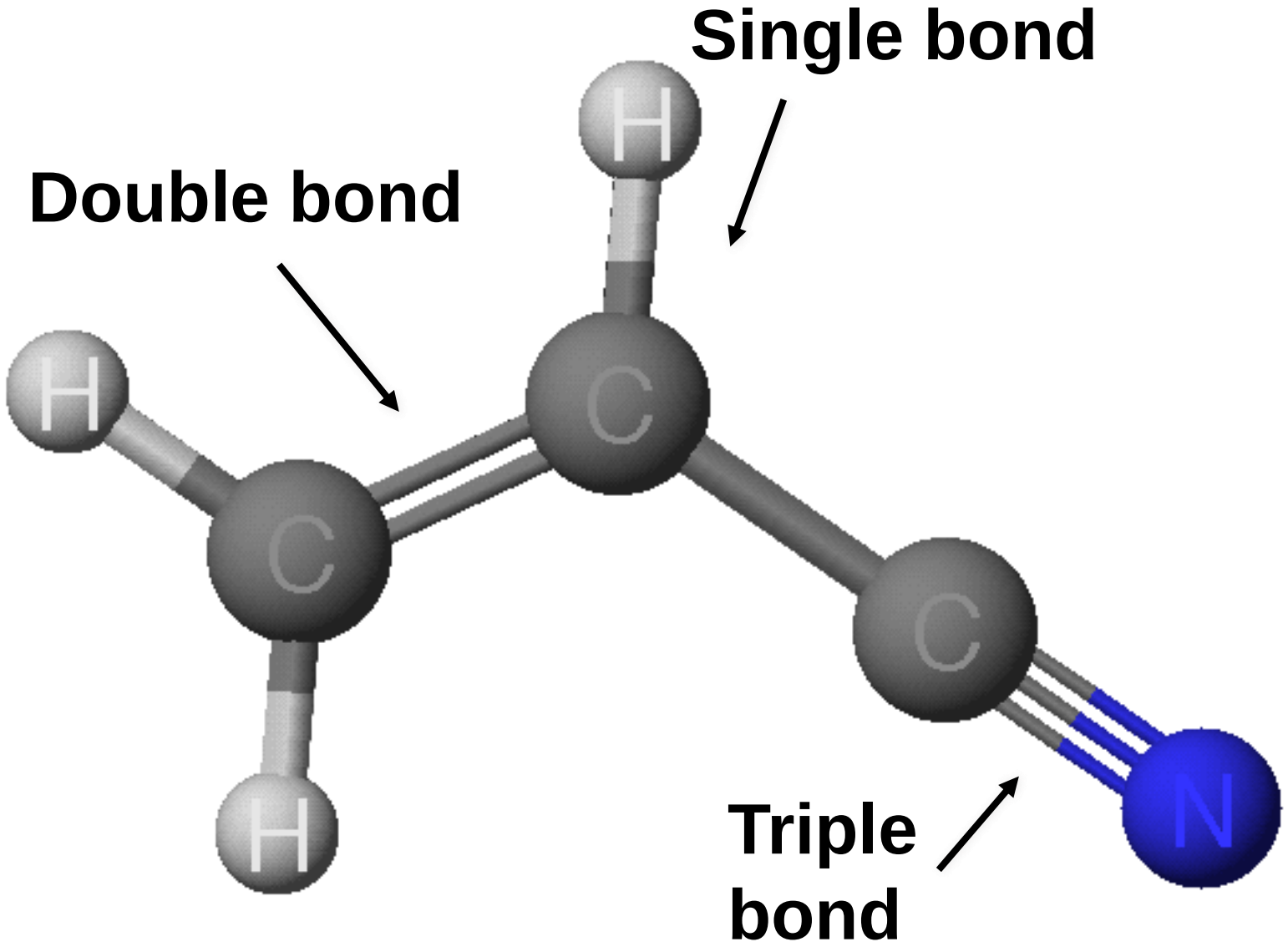
triple > *double* > *single*

Bond Order: 3 2 1

Bond length decreases with increasing bond order (Strength)

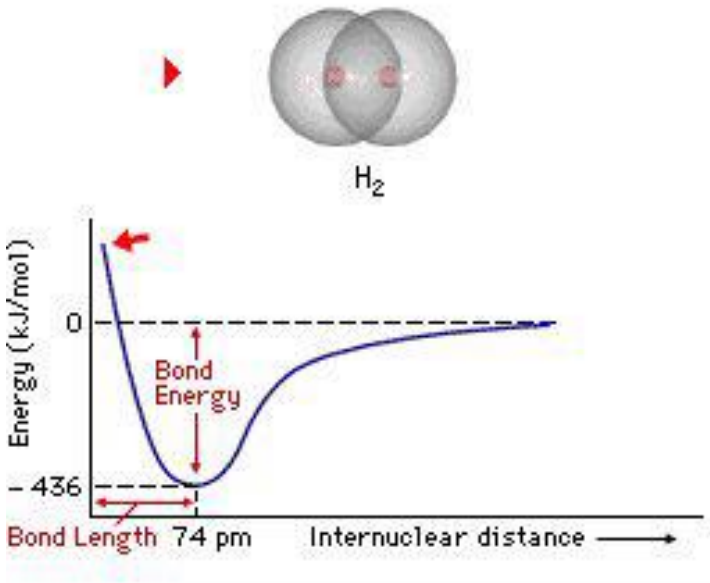
example: O=O bond is shorter than O–O bond

Bond Order: Number of Bonds Between a Pair of Atoms

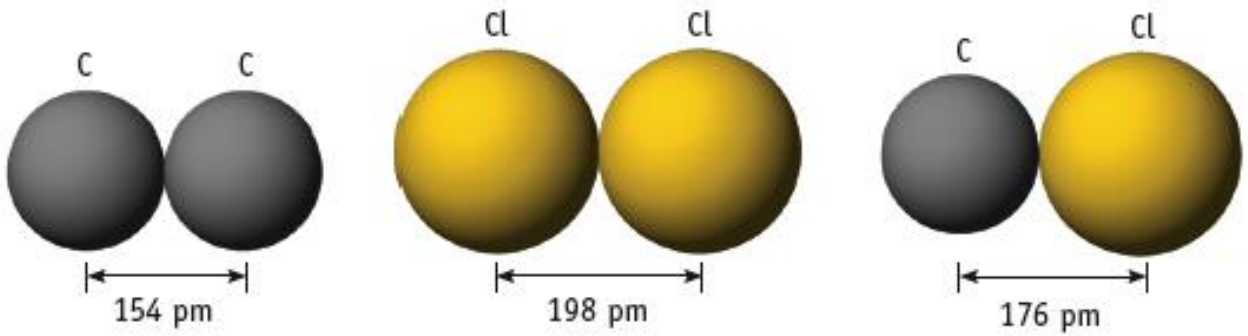


Acrylonitrile

Bond Length

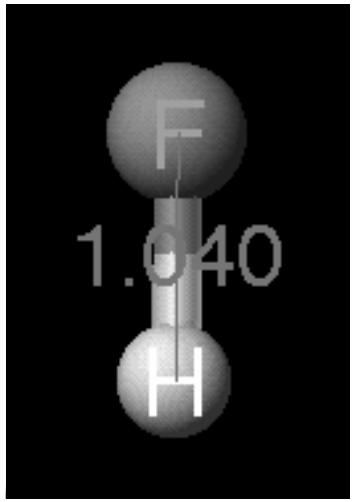


- Bond length is the distance between the nuclei of two bonded atoms.

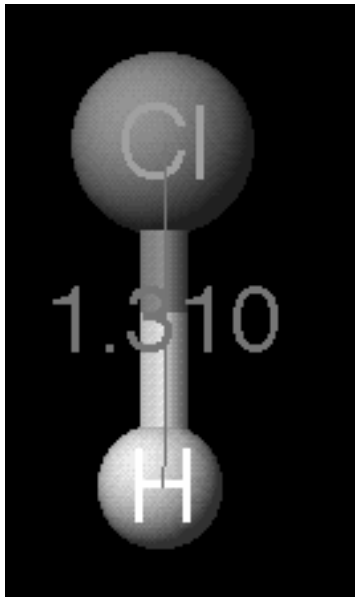


Bond Length

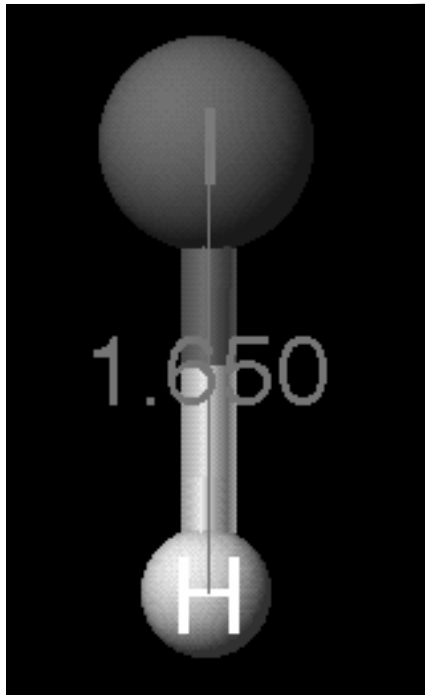
Bond length depends on the size of bonded atoms.



H—F



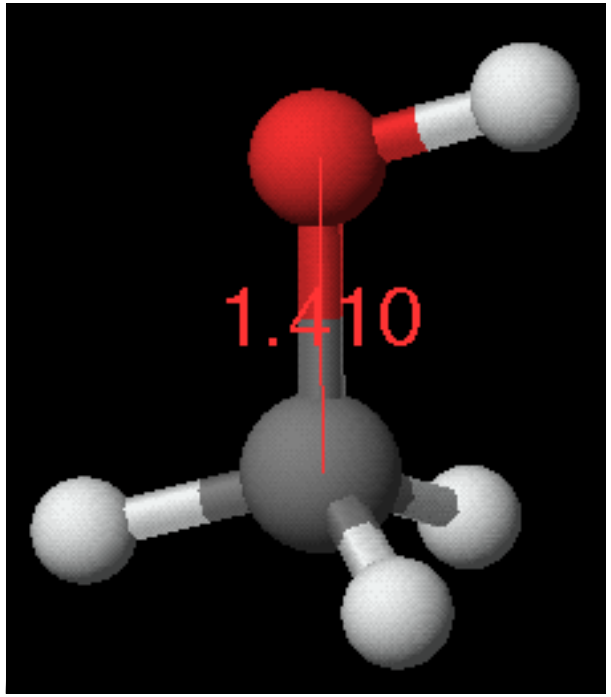
H—Cl

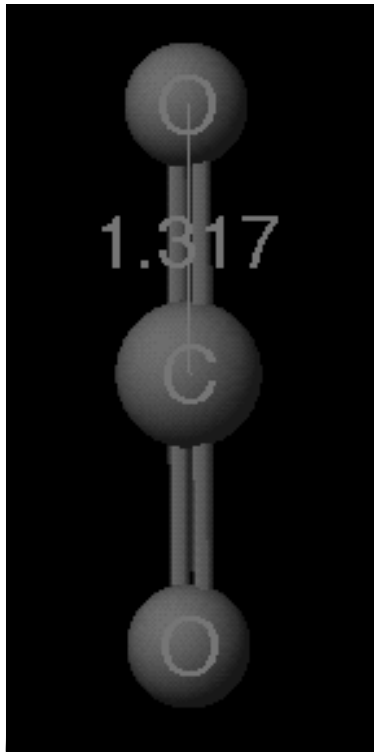


H—I

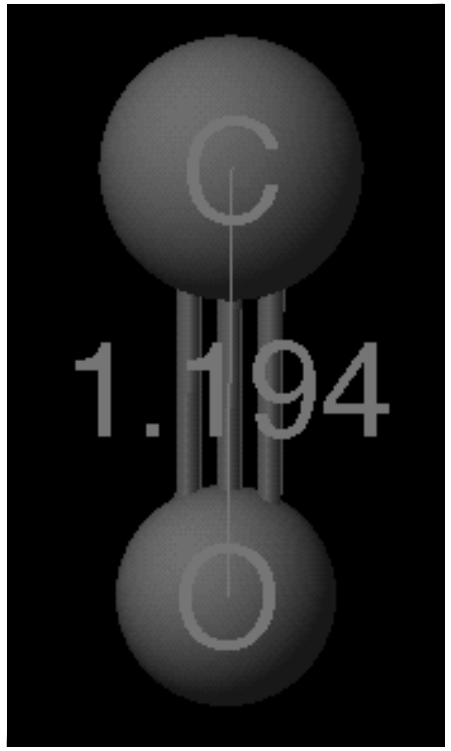
Bond distances measured in Angstrom units where
1Å = 10^{-10} m.

Bond Length





Bond length depends on bond order.



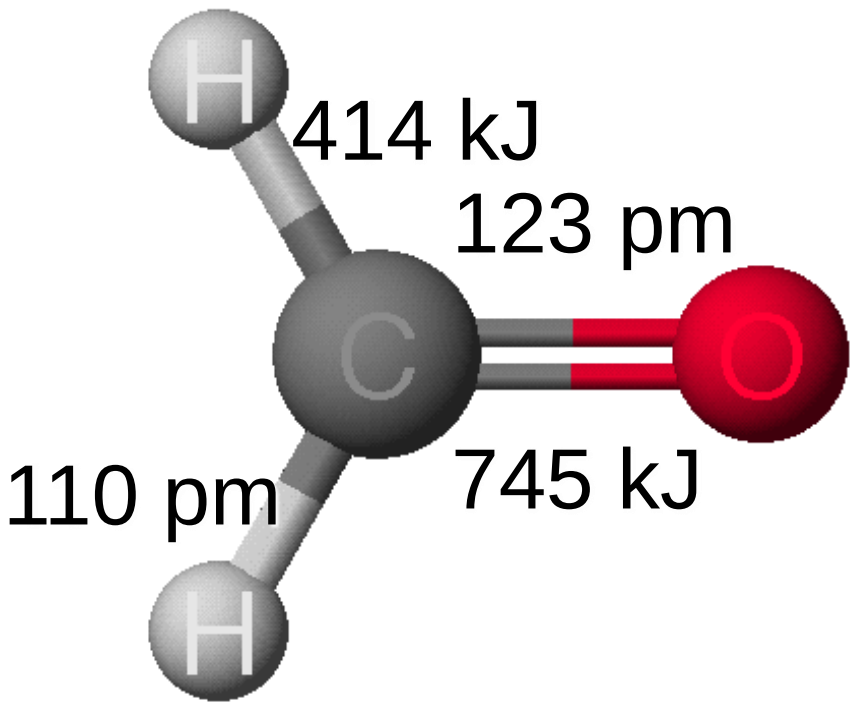
Bond distances measured in Angstrom units where
1Å = 10^{-10} m.

Bond Length & Bond Strength

Bond order is proportional to two important bond properties:

(a) bond strength

(b) bond length



Bond strength

- Bond enthalpy is the energy required to break a bond in a mole of gaseous substance.
  - $Cl–Cl(g) \rightarrow 2Cl(g)$ $\Delta H = D(Cl–Cl) = 242$ kJ
- Bond enthalpies are always +
  - energy is always required to break a bond
    - ΔH is always +
  - energy is always released when a bond forms
    - ΔH is always –
- Tabulated bond enthalpies are *averages*

Strengths of Covalent Bonds

The energy required to break a covalent bond is called the bond dissociation enthalpy, *D*.

For the Cl_2 molecule, *D*(Cl–Cl) is given by ΔH for the reaction:

$$Cl_2(g) \rightarrow 2Cl(g)$$

When more than one bond is broken:

$$CH_4(g) \rightarrow C(g) + 4H(g) \qquad \Delta H = 1660 \text{ kJ}$$

the bond enthalpy is a fraction of ΔH for the atomization reaction:

$$D(\text{C–H}) = \tfrac{1}{4}\Delta H = \tfrac{1}{4}(1660 \text{ kJ}) = 415 \text{ kJ}$$

Bond Dissociation Enthalpies

TABLE 8.8 Some Average Bond Dissociation Enthalpies (kJ/mol)*

Single Bonds											
	H	C	N	O	F	Si	P	S	Cl	Br	I
H	436	413	391	463	565	328	322	347	432	366	299
C		346	305	358	485	—	—	272	339	285	213
N			163	201	283	—	—	—	192	—	—
O				146	—	452	335	—	218	201	201
F					155	565	490	284	253	249	278
Si						222	—	293	381	310	234
P							201	—	326	—	184
S								226	255	—	—
Cl									242	216	208
Br										193	175
I											151

Multiple Bonds			
N=N	418	C=C	610
N≡N	945	C≡C	835
C=N	615	C=O	745
C≡N	887	C=O (in CO_2)	803
O=O (in O_2)	498	C≡O	1046

*Sources of dissociation enthalpies: I. Klotz and R. M. Rosenberg, *Chemical Thermodynamics,* 4th Ed., p. 55, New York, John Wiley, 1994; and J. E. Huheey, E. A. Keiter, and R. L. Keiter, *Inorganic Chemistry* 4th Ed., Table E. 1, New York, HarperCollins, 1993. See also Lange's *Handbook of Chemistry,* J. A. Dean (ed.), McGraw-Hill Inc., New York.

BOND	Bond dissociation enthalpy (kJ/mol)
H—H	436
C—C	346
C=C	602
C≡C	835
N≡N	945

The GREATER the number of bonds (bond order) the HIGHER the bond strength and the SHORTER the bond.

Using Bond Enthalpy to Calculate $\Delta_r H$

- $\Delta_r H = \Sigma(\text{bonds broken}) - \Sigma(\text{bonds formed})$

- $H_2 + Cl_2 \rightarrow 2HCl$

436 242 2 x 432

678 - 864= −186 kJ/ 2 mol HCl

So, $\Delta_f H$ (HCl) = -93 kJ/mol